# EXP 14

# Explore and Manipulate the ChickWeight Dataset

Aim:

To explore and manipulate the built-in `ChickWeight` dataset by sorting, melting, and casting the data according to Diet.

Procedure:

1. Load the `ChickWeight` dataset.
2. Display its structure and first few rows.
3. Sort the dataset by weight.
4. Melt the data into long format.
5. Cast the data to calculate average weight by Diet and Time.

Code:

```r
install.packages("reshape2", repos = "https://cloud.r-project.org")
library(reshape2)

data(ChickWeight)

print(head(ChickWeight))
print(str(ChickWeight))

sorted_data <- ChickWeight[
  order(ChickWeight$weight),
]

print(head(sorted_data))

melted_data <- melt(
  ChickWeight,
  id.vars = c("Chick", "Diet", "Time"),
  measure.vars = "weight"
)

print(head(melted_data))

cast_data <- dcast(
  ChickWeight,
  Diet ~ Time,
  value.var = "weight",
  fun.aggregate = mean
)

print(cast_data)
```

OUTPUT:

```
   weight Time Chick Diet
1      42    0     1    1
2      51    2     1    1
3      59    4     1    1
4      64    6     1    1
5      76    8     1    1
6      93   10     1    1
```

RESULT:

Thus, the `ChickWeight` dataset was explored, sorted, melted, and cast according to Diet and Time successfully.